Identifying Abnormal Prescribing Patterns in High-Risk Drugs

Caleb A. Fernandes

***Abstract*—High-risk medications such as opioids and benzodiazepines continue to drive adverse patient outcomes through misuse and overprescription. Identifying abnormal prescribing behavior within electronic health record (EHR) data is therefore essential for improving patient safety and guiding targeted interventions. In this work, we construct a weighted bipartite provider–drug network linking prescribers to high-risk drugs observed in the OMOP standardized EHR schema. Using network analysis techniques, we apply Personalized PageRank (PPR) and subgraph-level metrics—including weighted degree, bipartite density, and square clustering coefficient—to quantify both global prescribing structure and local deviations from specialty norms. Our analysis reveals a highly skewed, scale-free distribution of exposures in which a small subset of providers accounts for a disproportionate share of high-risk prescriptions. We identify one provider, termed *Provider Zero*, whose prescribing volume substantially exceeds that of peers within family medicine. Induced specialty subgraphs show that *Emergency Medicine* and *Anesthesia* exhibit cohesive, well-defined prescribing patterns, whereas *Residency* and *Nurse Practitioner* networks are more fragmented, indicating heterogeneity in medication use. These findings demonstrate how graph-based modeling can uncover hidden structure and outlier behavior in EHR data, offering a scalable framework for monitoring prescribing practices and assessing patient-safety risks across medical specialties.**

I. Introduction

Prescription drug misuse remains a critical public health issue in the United States, contributing to more than 100,000 overdose deaths annually according to recent CDC reports [1]. Among the primary drivers are high-risk drug classes such as opioids, benzodiazepines, and central nervous system (CNS) stimulants, which carry elevated risks of addiction, respiratory depression, and drug–drug interactions [2]. Monitoring and mitigating inappropriate prescribing practices is therefore essential to improving patient safety and reducing preventable morbidity. However, large-scale prescription monitoring remains challenging because prescribing behavior emerges from complex interactions between providers, specialties, and drugs—relationships that are difficult to interpret using traditional tabular or aggregate statistics.

Electronic Health Record (EHR) systems capture these interactions at scale, offering a unique opportunity to study prescribing behavior through network-based methods. Network analysis has become an increasingly powerful tool in healthcare for uncovering structural patterns across clinical entities. Prior studies have used graph-based approaches to model disease co-occurrence networks [3], temporal progression of diagnoses [4], patient–drug interaction networks [5], and physician referral or collaboration graphs [6]. These works demonstrate how complex healthcare systems exhibit community structure, centrality hierarchies, and non-random connectivity patterns that can reveal inefficiencies, risks, and hidden modes of care delivery. Despite this, the application of network analysis to *prescriber–drug relationships*—especially for identifying abnormal prescribing of high-risk medications—remains relatively unexplored.

In this study, we focus on detecting abnormal prescribing patterns of high-risk drugs across medical specialties using a bipartite provider–drug network derived from EHR data. Our central research question is: *Are prescribing rates for high-risk drugs contained within the normal range of their respective specialties, or do certain providers exhibit abnormal patterns that deviate from the norm?* To answer this, we construct a weighted bipartite graph linking providers to the high-risk drugs they prescribe and employ both global and local network analysis techniques. Specifically, we apply Personalized PageRank (PPR) to assess similarity among providers within and across specialties, compute weighted and unweighted degree distributions to capture prescribing intensity and diversity, and evaluate induced specialty subgraphs using bipartite density and square clustering coefficients to quantify prescribing cohesion.

Our contributions are threefold. First, we present a reproducible pipeline for constructing provider–drug networks from standardized EHR data using OMOP vocabulary mappings, enabling cross-specialty comparisons of prescribing behavior. Second, we integrate Personalized PageRank and bipartite graph metrics to identify both specialty-level trends and provider-level outliers, uncovering how certain providers—such as the identified *Provider Zero*—deviate from normative prescribing behavior. Finally, we provide patient-centered interpretations of these findings, highlighting how structural network cohesion corresponds to prescribing consistency and how fragmentation or abnormal centrality may signal potential patient safety risks. By combining network science with clinical domain reasoning, this work contributes to the growing field of data-driven pharmacovigilance and offers a scalable framework for monitoring prescription behavior within healthcare systems.

II. Network Construction Details

Our network consisted of a bipartite graph between providers and high-risk drugs. The edges between provider p and drug d were established if there was a patient who was prescribed drug d by provider p. Edges were weighted according to the number of times d was prescribed by p, including refills. The finalized graph in Figure 2 contains 123 unique drugs, 2,139 unique providers, and 4,270 edges.

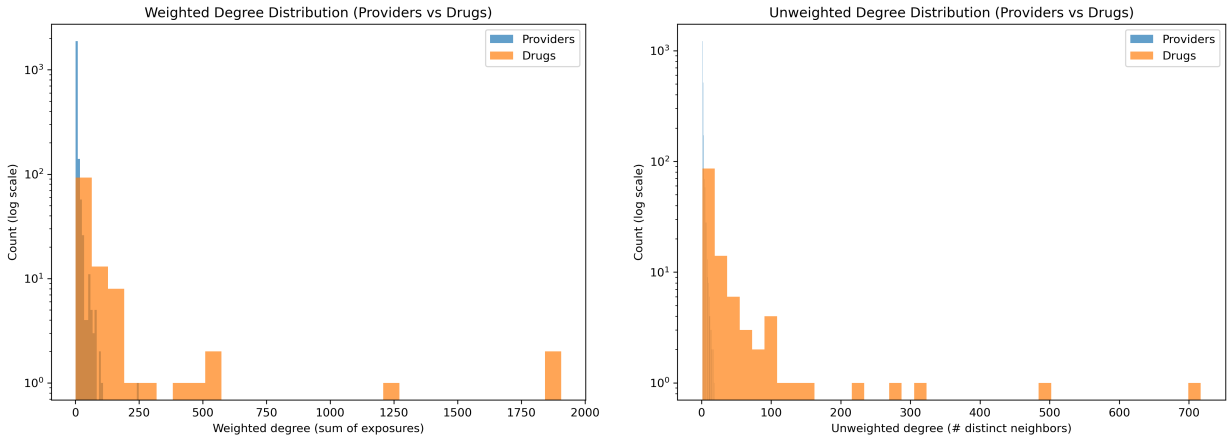


Fig. 1. (a) Weighted and (b) Unweighted Degree Distributions of Providers and Drugs

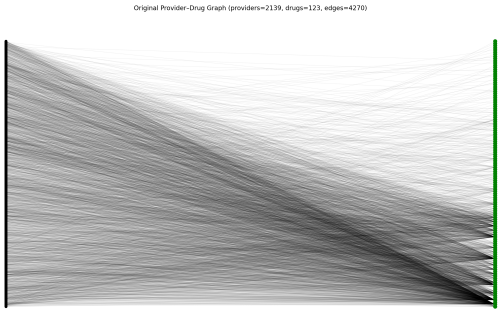


Fig. 2. Complete Bipartite Graph of Providers (left side) and Drugs (right side)

Drug Preprocessing

To isolate exposures to high-risk medications, we implemented a three-stage filtering process using the Observational Medical Outcomes Partnership (OMOP) vocabulary, ancestor, and drug exposure tables. We identified five high-risk drug ancestor concepts corresponding to major controlled-substance categories, consisting of opioid analgesics, benzodiazepine receptor agonists, and CNS stimulants. These ancestor identifiers were retrieved from the standardized concept hierarchy in the Athena OHDSI website https://athena.ohdsi.org/search-terms/start and stored as the reference set. We expanded these ancestors through the `concept_ancestor` table to include all descendant drug concepts, yielding a comprehensive list of 55,652 standardized drug identifiers. We filtered the `drug_exposure` table by retaining only records whose drug concept identifiers appeared in this descendant list. The resulting subset contained 58,549 valid exposures (approximately 7.5% of the original 777,234 rows) spanning 123 unique drug concepts. This pipeline ensured that subsequent analyses focused exclusively on clinically relevant high-risk prescriptions while preserving detailed exposure-level information such as provider linkage and exposure timing.

Provider Linking

From the provider table, we retained only `provider_id`, `specialty_concept_id`, and `specialty_source_value`. From the drug exposure file, we kept `person_id`, `drug_concept_id`, `provider_id`, and the exposure start and end dates, parsed dates to date times, coerced IDs to integers, and removed rows with missing keys. Importantly, to capture potential overprescription behavior, we did not collapse refills: repeated exposures for the same patient–provider–drug combination were preserved as independent records. Finally, we restricted the provider table to prescribers observed in the drug exposure table and filtered out ”Unknown” specialty providers by removing providers who did not have values in their `specialty_source_value`. We merged the high-risk drug exposure records with provider information to attach each exposure to the provider’s specialty. We perform an inner join on `provider_id`, keeping only exposures linked to known providers with valid specialties. The output writes the cleaned and merged dataset to `high_risk_drug_exposure_with_providers.csv`. The 7,262 unique exposures contained 2,139 unique providers. The most represented specialties by exposure were Emergency Medicine (2,718), Anesthesia (1,815), Nurse Practitioner (1,370), and Internal Medicine—patterns (973). The final provider-linked dataset was saved and serves as the foundation for constructing the provider–drug bipartite network.

Graph Construction

Using the output file from the last step, we aggregated the data by each unique `(provider_id, drug_concept_id)` pair, counting both the total number of exposures (`n_exposures`) and the number of unique patients (`n_patients`) associated with that pair. Using this aggregated table, a bipartite graph was built in NetworkX where provider nodes are labeled with their specialty, drug nodes represent unique drug concepts, and edges connect providers to drugs they have prescribed. Each edge is weighted by the total number of exposures and annotated with the corresponding number of patients. The resulting graph thus encodes

the full provider–drug prescribing relationships, forming the structural foundation for subsequent network analyses.

Toy Graph Construction and Visualization

To provide a better visual of the network, a small representative bipartite graph was generated by sampling 30 providers and 10 drugs from the real network, as seen in Figure 3. This toy graph served solely as a visualization tool and not for metric computation. Provider nodes were colored by specialty (excluding green to avoid confusion with drugs), while drug nodes were rendered in green. Edge color and width were scaled by exposure count, with red gradients denoting higher prescription volume.

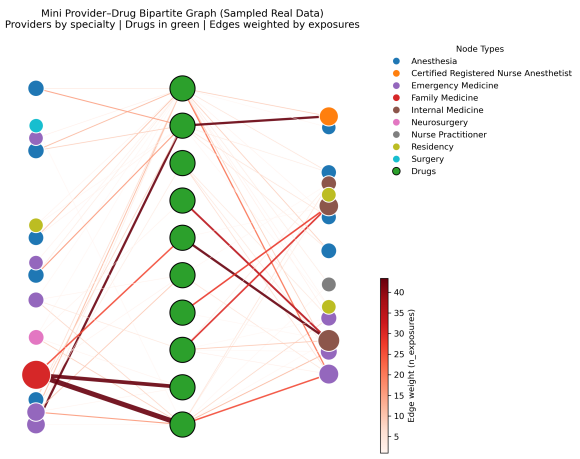


Fig. 3. Toy Graph Visualization

III. Metrics and Analysis Methods

As part of our analysis, we apply Personalized PageRank (PPR) and compute network measures—weighted degree (exposure count), unweighted degree, square clustering coefficient, density of induced specialty subgraphs, number of disconnected components, and the giant-component fraction—on the full graph and on specialty-specific induced subgraphs. Our research question asks whether providers within the same specialty exhibit similar prescribing patterns and how to flag departures from those norms; each metric is chosen to address one facet of that question. PPR, seeded at top providers within a specialty, tests whether random walks preferentially remain inside that specialty (specialty coherence). Weighted degree quantifies prescribing intensity so the top provider can be compared to the specialty mean (deviation detection), while unweighted degree captures the breadth of drug/provider connections (prescribing scope). The square clustering coefficient measures co-prescription "squares" in our bipartite network (local consensus), density of specialty-induced subgraphs captures global cohesion (within-specialty similarity), and the counts of disconnected components together with the giant-component fraction reveal fragmentation inconsistent with homogeneous practice (potential anomalies). Together these metrics let us verify specialty coherence, highlight unusually influential providers, and prioritize specialties for deeper investigation.

Global Metrics

Before performing PPR and metrics on induced subgraphs, we collected data on our base graph. Weighted degree is exposure count, and the unweighted degree represents unique drugs or providers linked to the node. Figure 1a and Figure 1b display the exposure count and unweighted degree of the nodes in the graph condensed to a log scale, respectively. Figures 4c and 4d present a pie chart showing the distribution of the specialty composition by the top 5% exposures and top 5% drugs prescribed. From this, we develop the scatter plot seen in Figure 5 where we plot all the providers, highlighting the top 5% exposure providers and top 5% drug prescribing providers. Finally, we subset the graph by provider specialty and compare the highest exposure provider to the mean exposure rate of their specialty. We do this again for the prescribed drug count. The heatmap in Figure 6a provides the results for the top 7 specialties and their respective comparisons. The top 7 specialties are defined by the top 7 specialties with the highest ratio of exposure to specialty mean. Intuitively, providers of the same specialty should have the same prescribing behavior - same exposure counts and number of drugs prescribed. Figure 6a provides a set of specialties to begin analyzing prescribing behavior.

Personalized Page Rank

We chose PPR under the heuristic assumption that providers who prescribe the same drug should be of the same specialty. For an extreme example, if an emergency medicine provider is prescribing high dosages of opioids to subdue a burn victim, that same prescription should not be being frequently prescribed by a family medicine provider. To accommodate the rare case where this could happen, we weighted PPR by edge weight where the walk is more likely to take edges with higher weights. This captures providers who are frequently providing that drug.

We ran PPR on the top provider of the top 7 specialties identified in Figure 6a to take a deeper dive into the abnormal prescribing behavior of these specialties. In total, 7 iterations of PPR was run where the restart node was the top provider of the respective specialty for which the run was designated to. Figure 6b provides the heatmap of our results. The vertical axis is the top provider of the respective specialty the corresponding boxes indicate the frequency in which that specialty was recorded during PPR. We intentionally only recorded the top 7 specialties for the sake of readability. For a more comprehensive study, all the specialties should be listed on the horizontal axis.

Provider Zero

Provider Zero is the name attributed to the outlier provider seen in Figure 5 who is family medicine yet has the highest exposure rate and drugs prescribed. Provider Zero is also seen in Figure 6a as the highest ratio of exposures in relation to

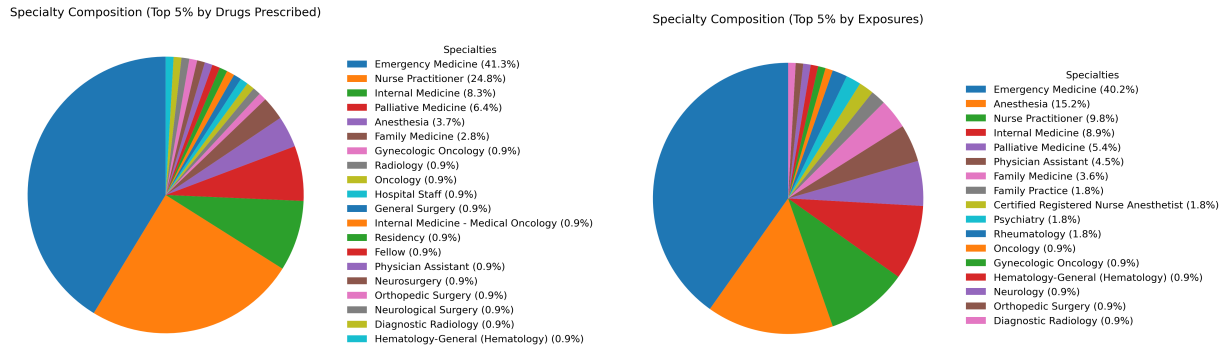


Fig. 4. Composition of Specialties for the Top 5% of Providers by (a) Drugs Prescribed and (b) Exposure

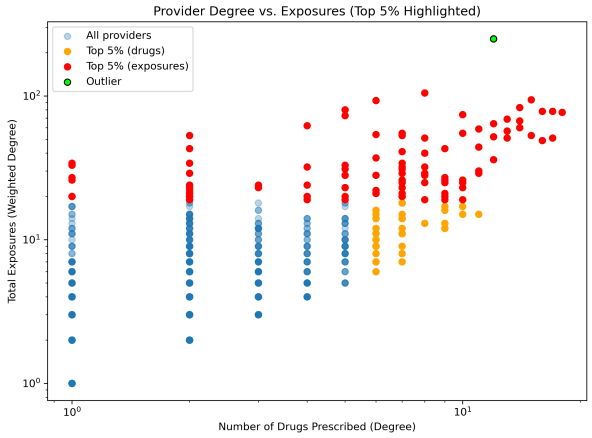


Fig. 5. Distribution of Exposure Counts and Drugs Prescribed Highlighted by the Top 5% in each Metric

mean. In order to study their behavior, we constructed an induced subgraph of Provider Zero, seen in Figure 7. This 2-hop subgraph aims to complement the PPR derived on the top provider of each specialty. We label the top specialties present in the 2-hop subgraph as well as the names of the top drugs prescribed by Provider Zero, thus allowing us to compare local structure in the 2-hop subgraph to the global influence seen in PPR.

Induced Subgraphs

Our final analysis provided insights into the top specialties by provider count. First, we identified the top 5 specialties by provider count. Our results were Residency, Anesthesia, Nurse Practitioner, Emergency Medicine, and Physician Assistant in that order. Then, we created 5 induced subgraphs of all the providers of each specialty and their respective induced drugs. On each of the 5 graphs, we calculated the number of providers, number of unique drugs, number of edges, density, mean square clustering coefficient, number of disconnected components, and the fraction of the giant component in relation to the total number of nodes. Since our graph is a bipartite graph, the traditional clustering coefficient and density formula cannot be used. Instead, we use the square clustering coefficient, which describes a cluster in a bipartite graph as a square rather than the traditional triangle, which is impossible in a bipartite graph. For density, unlike a traditional undirected graph, where edges can exist between any pair of nodes, a bipartite graph only allows edges between nodes of different types (e.g., providers and drugs). Consequently, the maximum number of possible edges is the product of the sizes of the two partitions. Table 1 provides the results for this experiment.

For a given node i, the local square clustering coefficient is defined as:

$$C_i^{\square} = \frac{2S_i}{k_i(k_i-1) - 2T_i} \tag{1}$$

where:

- $C_i^{\square}$ — square clustering coefficient of node i,
- S_i — number of distinct squares (4-cycles) that include node i,
- k_i — degree of node i,
- T_i — number of triangles that include node i, subtracted to avoid overcounting.

The global square clustering coefficient averages the local coefficients across all nodes in the graph:

$$C_{\text{global}}^{\square} = \frac{1}{N}\sum_{i=1}^{N} C_i^{\square} \tag{2}$$

where:

- $C_{\text{global}}^{\square}$ — global (mean) square clustering coefficient of the network,
- N — total number of nodes in the graph,
- $C_i^{\square}$ — local square clustering coefficient for node i.

For a bipartite graph $G = (U, V, E)$ with partitions U and V, the density is defined as:

$$\text{Density}_{\text{bip}}(G) = \frac{m}{|U|\,|V|} \tag{3}$$

where:

- $m = |E|$ — total number of observed edges in the graph,
- $|U|$ — number of nodes in the first partition (e.g., providers),
- $|V|$ — number of nodes in the second partition (e.g., drugs).

Intuitively, we should be seeing high density, clustering coefficient, and a low number of disconnected components. This

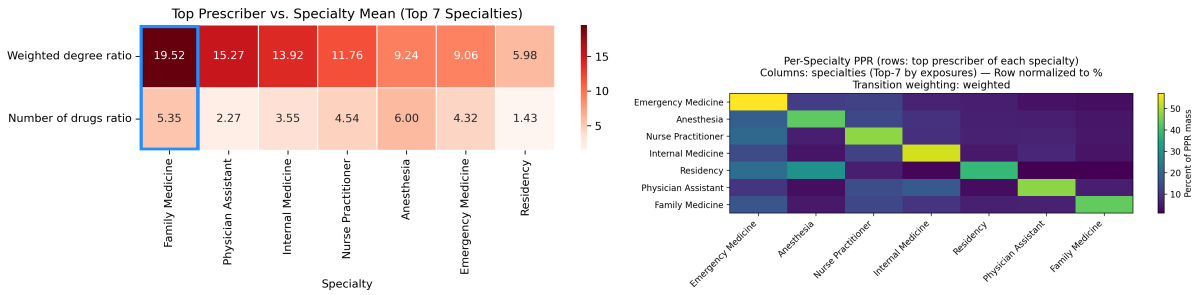


Fig. 6. Analysis of Top 7 Providers. (a) Comparison of mean exposure and drug count and (b) PPR results beginning from each provider

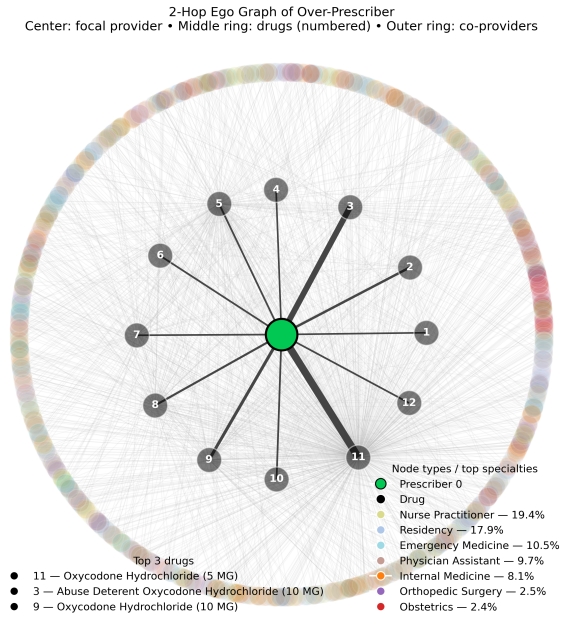


Fig. 7. 2-Hop Induced Subgraph of Provider Zero

operates under the belief that providers of the same specialty, when providing the same drug, produce squares captured in the clustering coefficient. This contributes towards validating the drug as being "normally" prescribed by this specialty on a local scale. On a global scale, providing the same set of drugs would mean little abnormal behavior and would be represented in high density. Disconnected components would be a sign that there are singleton or small communities of providers prescribing different drugs from the giant component. Operating under the belief that providers of the same specialty should be prescribing the same drugs, a large number of these communities could suggest that there are providers within this specialty that are prescribing drugs not native to that specialty. This increases the risk of patients to be exposed to high risk drugs that are not common to a specialty. To bring more light into the disconnected components, we compute the fraction of the giant component in relation to the whole subgraph to put the number of disconnected communities in more context.

IV. Interpretations and Conclusions

This section interprets the network structures we observed in the context of clinical practice and patient safety. Broadly, our results indicate that specialties with high subgraph density, strong square clustering, and a dominant giant component reflect cohesive, specialty-consistent prescribing (e.g., *Emergency Medicine*, *Anesthesia*), whereas fragmented subgraphs signal heterogeneous, potentially inconsistent practice within a specialty. Personalized PageRank (PPR) seeded at top providers largely remains within specialty boundaries, supporting the notion of "specialty coherence," while deviations in weighted and unweighted degree highlight unusually influential or wide-scope prescribers. We use these structural cues to (i) identify abnormal behavior at both the specialty level and the provider level (with a focused analysis of *Provider Zero*), (ii) discuss implications for patient risk and care coordination, and (iii) delineate limitations such as representation bias and lack of visit-level context. We close by outlining future directions, including scaling to the full dataset, adding richer event types and demographics, and exploring GNN-based models to move from descriptive signals to predictive monitoring.

Identifying Abnormal Behavior

The bulk of our analysis focused on abnormal prescribing within specialties. To contextualize those findings, we first examined global metrics. Figure 4b shows the specialty composition among the top 5% of providers by exposure; *Emergency Medicine* dominates, followed by *Anesthesia* and *Nurse Practitioner*. This aligns with Figure 4a, where *Emergency Medicine* also exhibits the largest number of unique drugs, followed by *Nurse Practitioner*, *Internal Medicine*, and others. These patterns are clinically intuitive: emergency clinicians face a wide range of conditions requiring immediate intervention [**?**], so they both prescribe frequently and draw from a broad drug pool. By contrast, anesthesiologists use a narrower, specialized set of drugs very frequently—placing them lower on unique-drug breadth but higher on exposure counts. Overall, the global distributions are consistent with expected practice norms across specialties.

Global analysis also revealed clear anomalies. The degree distributions for providers and drugs (Figure 1a) are heavy-tailed: most nodes have near-zero exposure counts with a long right tail. Within this tail, we observe an outlier provider with ~ 250 exposures and a drug exceeding $1{,}750$ exposures—both

TABLE I
TOP-5 SPECIALTY SUBGRAPHS — DENSITY, GLOBAL SQUARE CLUSTERING, AND COMPONENTS

Specialty	Providers	Drugs	Edges	Density	Mean Sq. CC	Components	Largest Comp. Frac
Residency	370	48	519	0.029	0.598	10	0.955
Anesthesia	305	44	508	0.038	0.612	3	0.986
Nurse Practitioner	273	71	661	0.034	0.296	6	0.971
Emergency Medicine	262	55	909	0.063	0.252	2	0.994
Physician Assistant	131	40	231	0.044	0.384	5	0.953

far beyond the bulk. The unweighted degree distribution (Figure 1b) is similarly heavy-tailed, with a few drugs connected to an unusually large number of unique providers. This pattern is visually corroborated in Figure 2, where bottom-right drug nodes exhibit disproportionately more edges. Together, these signals motivate a closer examination of abnormal prescribing behavior.

To refine our search for a focal case, we plotted all providers on logarithmic axes (Figure 5) and highlighted those in the top 5% by exposures and by number of drugs prescribed. One provider stands out sharply from the field—marked in green—with the highest exposure count relative to drugs prescribed. We refer to this provider as *Provider Zero*. Notably, *Provider Zero* belongs to *Family Medicine*, a specialty that does *not* appear among the dominant contributors to exposure or drug-count composition in Figure 4b. Thus, we have identified a provider from a globally underrepresented specialty exhibiting abnormal prescribing behavior.

Building on *Provider Zero*'s atypical prescribing, we compared each specialty's top provider to that specialty's mean exposure and drug count. The top seven specialties by exposure ratio are shown in Figure 6a. *Provider Zero* exhibits the highest ratio; however, Figure 6a also indicates that top providers in several specialties deviate substantially from their respective means. These results provide (i) clear evidence of abnormal prescribing in the network, (ii) a concrete focal case for deeper analysis, and (iii) a comparative basis for examining additional specialties.

Analyzing Top Specialties By Exposure

We performed weighted PPR on the seven specialties with the highest exposure ratios identified in Figure 6a. For each specialty, we ran an independent PPR with the restart node set to that specialty's top provider (seven runs total). Figure 6b summarizes the specialties encountered along each random walk, providing a global view of top-provider behavior. If providers largely operate within their domain, the heatmap should be diagonally dominant—i.e., during a walk seeded in specialty s, nodes from s should appear most frequently. Because PPR is weighted by exposure, it de-emphasizes singletons and amplifies highly exposed ties. Consistent with this expectation, the rows (horizontally) in Figure 6b show strong diagonal peaks across specialties. Interpreting the columns (vertically), *Anesthesia* and *Family Medicine* present the most contrastive patterns: anesthesia rarely appears along paths seeded in other specialties (except *Residency*), which is reassuring if those residency providers are anesthesiology trainees; otherwise it may warrant attention given anesthesia's higher-risk pharmacologic profile. Family medicine also appears infrequently along other specialties' walks, which could indicate (i) a coherent family-medicine subcommunity prescribing a distinct set of drugs, or (ii) under-representation of family-medicine nodes limiting global visibility.

The PPR patterns and subgraph structure have direct implications for how patients experience care and how clinicians relate within and across specialties. Diagonal dominance in Figure 6b suggests strong "within-specialty" prescribing coherence, which aligns with standardized protocols and can support predictable, guideline-concordant treatment pathways for patients. The relative isolation of *Anesthesia* from other specialties' walks is consistent with concentrated expertise in higher-risk medications, implying that cross-specialty exposure to those drugs is limited—potentially reducing risk. Conversely, the sparse visibility of *Family Medicine* along other specialties' paths may reflect either a healthy differentiation of primary-care pharmacotherapy or under-representation that can obscure care coordination needs.

Analyzing Provider Zero

We conducted a local analysis of *Provider Zero* by constructing an induced subgraph of the provider's 2-hop neighborhood, capturing both the drugs they prescribe and the other providers who also prescribe those drugs. Figure 7 visualizes this ego network, with thicker edges indicating higher exposure counts and the legend summarizing neighbor specialty composition. Locally, the majority of neighbors are *Nurse Practitioner* (19.4%), *Residency* (17.9%), and *Emergency Medicine* (10.5%). However, *Provider Zero*'s PPR does not show strong concentrations of these specialties, reinforcing our hypothesis that family medicine may be under-represented in the global graph and that other specialties are comparatively over-represented. Setting representation aside, the 2-hop view suggests that *Provider Zero* is adjacent to many cross-specialty providers, yet those providers do not exhibit high exposure to the same drugs. In contrast, the smaller set of family medicine neighbors shows stronger exposure links to *Provider Zero*'s drug set. This explains the visible ring of non–family medicine providers in the local subgraph that diminishes under global PPR, where exposure weighting and specialty composition reduce their prominence.

We further examined *Provider Zero*'s drug mix to contextualize their high exposure rate. The top three drugs are *Oxycodone Hydrochloride (5 MG)*, *Abuse-Deterrent Oxycodone Hydrochloride (10 MG)*, and *Oxycodone Hydrochloride (10 MG)*. One patient accounts for over 100 exposures—nearly

half of *Provider Zero*'s ~ 250 total. So, without visit-level context, we refrain from condemning this provider. Notably, the prominence of abuse-deterrent oxycodone suggests an attempt to mitigate misuse risk, and the higher frequency of 5 mg relative to 10 mg doses is consistent with conservative initiation or dose-tapering practices.

At the individual level, findings around *Provider Zero* underline the importance of monitoring high-exposure outliers while preserving clinical nuance: dose patterns (e.g., 5 mg vs. 10 mg oxycodone, abuse-deterrent formulations) can signal risk mitigation or tapering strategies rather than misuse. Operationally, these network cues can inform targeted chart reviews, strengthen referral and co-management pathways (e.g., pain management, addiction medicine), and prioritize stewardship efforts where fragmentation or atypical cross-specialty ties could elevate patient risk.

Analyzing Induced Subgraphs

To get a more complete picture of the most represented providers in the dataset by raw provider count, we created induced subgraphs of these providers and calculated metrics seen in Table 1. These results confirm what we suggested earlier about a discrepancy in specialty representation in the graph. The specialties seen in Table 1 are frequently recorded in our other studies, this could be because their specialties have a high mean exposure or drug prescribing rates, or because they have more representation in the data. The results in Table I provide a structural overview of the induced subgraphs for the top five specialties by provider count. Among these, *Emergency Medicine* and *Anesthesia* exhibit the highest densities and largest connected components, indicating that providers in these fields share highly overlapping prescribing behaviors and form cohesive sub networks. Both also have largest component fractions near one, meaning nearly all providers and drugs belong to a single dominant cluster with minimal isolation. In contrast, *Residency* and *Nurse Practitioner* subgraphs contain a greater number of disconnected components and smaller largest component fractions, suggesting fragmented prescribing structures or varying access to drug categories among providers. The higher number of components in *Residency*, in particular, may reflect the diversity of clinical rotations and supervision under different attending specialties. Meanwhile, *Physician Assistant* networks show moderate density and clustering but several smaller components, indicating intermediate cohesion. Overall, specialties with high density, strong clustering, and a single dominant component demonstrate homogeneity in prescribing patterns, whereas those with many disconnected components or lower largest component fractions reflect more distributed or variable prescribing behaviors within the specialty.

Implications for Patients

The network-level findings have meaningful implications for patients, particularly in understanding prescribing safety, consistency of care, and the potential risks arising from fragmented or atypical prescribing behaviors. Specialties such as *Emergency Medicine* and *Anesthesia*, which demonstrated high density, strong clustering, and large connected components, suggest that patients within these domains are likely to receive medications that are standardized and consistent with specialty-specific protocols. This structural cohesion reflects well-defined clinical practices where prescribers frequently interact with the same or similar sets of drugs, reducing the likelihood of inappropriate or off-domain prescribing. Such tightly connected prescribing networks are often associated with lower patient risk and clearer expectations for treatment safety.

In contrast, specialties like *Nurse Practitioner*, *Residency*, and *Physician Assistant*, which exhibit more fragmented subgraphs with multiple disconnected components, may indicate higher variability in prescribing behavior. For patients, this heterogeneity can translate into differences in medication exposure across practitioners of the same specialty—raising concerns about inconsistent treatment practices or potential over-reliance on certain high-risk drugs. The greater number of disconnected components within the *Residency* network, for example, may mirror varying levels of supervision, training environments, and specialty rotations, each of which can influence prescribing decisions and medication familiarity. While such variation is expected in educational contexts, it also highlights a need for strong oversight and standardized prescribing guidelines to mitigate risks to patients.

The identification of outliers such as *Provider Zero* underscores the importance of monitoring individual-level deviations within larger networks. A single provider prescribing far beyond the norm of their specialty introduces a measurable point of patient risk, whether through over-prescription, insufficient cross-specialty consultation, or exposure to medications that are not typically handled within that field. From a patient-safety perspective, such findings reveal how network analytics can serve as an early warning system—detecting atypical prescribing patterns before they manifest in adverse outcomes. Ultimately, these results emphasize that structural cohesion within a specialty is a proxy for prescribing consistency, and that both over-connected and under-connected regions of the network warrant attention: the former for potential systemic overuse, and the latter for gaps in prescribing oversight or coordination of care.

Limitations

Our work constrained PPR and subgraph metrics to ”top” specialties for the sake of simplicity. For a more comprehensive study, the methods described in this paper should be applied to all specialties. Furthermore, the number of providers in each specialty plays a critical role in identifying abnormalities. Having an equal representation of providers in each specialty makes our results far more robust since they are not biased by a higher number of nodes in certain specialties. It would be more beneficial to generate a larger volume of data such that there is a sufficient number of providers for each specialty. Defining abnormality is also significant in this study. For example, Provider Zero is labeled as an abnormal provider when looking purely at their ratio of exposures to the mean of family medicine. However, upon further analysis, it can be

seen that 100 of those exposures are from one unique patient. The details surrounding this patient are not discussed in this study and would require further research into their patient visits. Given the benefit of the doubt, it could be argued that this patient needed this high volume of prescriptions. As such, it is difficult to solely use this study and, in a binary fashion, label providers are abnormal or not. Instead, it highlights cases and specialties that can be further studied.

Future Directions

Future work would be to test this analysis on a larger scale. Our current data is a subset of a larger dataset. To get more accurate results, this would have to be tested on the full dataset. The research can be expanded in many directions. Using the patient demographic information, we could analyze if abnormal prescribing is correlated to a particular demographic - perhaps unveiling bias or prejudice in prescriber behavior. By gathering more information about the patients and visits, we could analyze which conditions led to higher prescribing of high-risk drugs. From there, we could uncover scenarios which are out of the norm - patients with low-risk conditions being prescribed high-risk drugs. We could use the data on death rates to uncover if there is a any correlation between deaths and abnormal prescribing behavior. This could be uncovered by linking deaths to drugs that were being taken when the patient died. In the context of machine learning, we could perform a binary classification task on a set of unidentified drugs to identify high-risk drugs based on prescription behavior of providers and their respective features.

REFERENCES

[1] C. for Disease Control and Prevention, "Provisional drug overdose death counts, united states, 2024," *National Center for Health Statistics*, 2024. [Online]. Available: https://www.cdc.gov/nchs/nvss/vsrr/drug-overdose-data.htm

[2] N. D. Volkow and F. S. Collins, "The evolving opioid epidemic: From prescription to addiction," *The New England Journal of Medicine*, vol. 384, no. 7, pp. 581–585, 2021.

[3] X. Zhou, J. Menche, A.-L. Barabási, and A. Sharma, "Human symptoms–disease network," *Nature Communications*, vol. 5, p. 4212, 2014.

[4] E. Choi, M. T. Bahadori, and J. Sun, "Gram: Graph-based attention model for healthcare representation learning," in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 787–795.

[5] Y. Zhang, Q. Yang, B. Zhao, and L. Chen, "Patient–drug graph learning for personalized medicine," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 29, no. 8, pp. 3554–3566, 2018.

[6] M. L. Barnett, N. A. Christakis, A. J. O'Malley, J.-P. Onnela, N. L. Keating, and B. E. Landon, "Physician networks and the costs of care in us hospitals," *The New England Journal of Medicine*, vol. 365, no. 19, pp. 1860–1869, 2011.